12.206

AI25BTECH11024 - Pratyush Panda

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Question:

The eigenvectors of the matrix

$$\begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \tag{0.1}$$

are written in the form $\begin{pmatrix} 1 \\ a \end{pmatrix}$ and $\begin{pmatrix} 1 \\ b \end{pmatrix}$. What is $a + b$?

a 0

b $\frac{1}{2}$

c 1

d 2

Solution:
Given

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \tag{0.2}$$

We have to find the eigen vectors. For that we first have to find the eigen values using the characteristic equation

$$det(\mathbf{A} - \lambda \mathbf{I}) = 0 \tag{0.3}$$

$$(1 - \lambda)(2 - \lambda) = 0 \tag{0.4}$$

$$\tag{0.5}$$

Hence,

$$\lambda_1 = 1, \quad \lambda_2 = 2 \tag{0.6}$$

Now, to find the eigenvector for each eigen value;

$$\mathbf{Av} = \lambda \mathbf{v} \tag{0.7}$$

$$\implies (\mathbf{A} - \lambda I)\mathbf{v} = 0 \tag{0.8}$$

$$\begin{pmatrix} 1-\lambda & 2 \\ 0 & 2-\lambda \end{pmatrix} \mathbf{v} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{0.9}$$

Writing as an augmented matrix and row-reducing:

$$\left(\begin{array}{cc|c} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \end{array}\right) \tag{0.10}$$

For $\lambda = 1$ we have;

$$\left(\begin{array}{cc|c} 0 & 2 & 0 \\ 0 & 1 & 0 \end{array}\right) \tag{0.11}$$

On solving, we get $\mathbf{v} = \begin{pmatrix} x \\ 0 \end{pmatrix}$.

Thus, the first eigen vector is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

For $\lambda = 2$ we have;

$$\left(\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \end{array}\right) \tag{0.12}$$

On solving this, we get $\mathbf{v} = \begin{pmatrix} x \\ \frac{x}{2} \end{pmatrix}$.

Thus, the second eigen vector is $\begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}$.

Thus, when we get $a = 0$ and $b = \frac{1}{2}$. So $a + b = \frac{1}{2}$. Therefore, option d is correct.